

Specific heat and thermal diffusivity of hardening concrete

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This paper focuses on the evolution of the thermal characteristics of early-age concrete as a function of the state of the hardening process. A literature review is presented, together with new test results obtained by means of improved test methods: For the determination of the specific heat of cement paste during hardening and of the thermal diffusivity of hardening concrete, test methods are developed based on existing methods. The specific heat is determined for hardening cement paste samples made with blastfurnace slag cement. It is concluded that the specific heat and thermal diffusivity decreases linearly with the degree of hydration.

Notation

- a thermal diffusivity of hardening concrete
- c specific heat of hardening concrete
- c_0 specific heat of hardened concrete
- c_a specific heat of aggregates
- c_c specific heat of cement
- c_{cf} fictitious specific heat of hydrated part of cement
- c_p specific heat of cement paste
- c_w specific heat of water
- E_1 energy supply in the absence of the cement paste
- E_2 energy supply in the presence of the cement paste
- G_a amount by weight of aggregate
- G_c amount by weight of cement
- G_w amount by weight of water
- m_p mass of the cement paste sample
- r degree of hydration
- $\Delta\theta$ initial temperature difference between concrete and waterbath (diffusivity test)
- $\Delta\theta_1$ temperature increase due to E_1
- $\Delta\theta_2$ temperature increase due to E_2

- θ temperature
- θ_0 initial temperature of concrete (diffusivity test)
- λ thermal conductivity of hardening concrete
- ρ density of concrete

Introduction

At the University of Ghent, several research programmes have been executed concerning massive concrete armour units used for breakwaters. In particular, cracking due to thermal stresses caused by the heat of hydration, and impact resistance were studied.^{1–3} Due to these thermal stresses external as well as internal crack formation can occur during hardening. Wave loads can stimulate crack growth. Thus, service life can be reduced markedly by thermal cracks. A thorough knowledge of the thermal stresses and crack formation is necessary in order to evaluate their influence on durability. At the University of Ghent a finite element program for the prediction of thermal stresses due to the heat of hydration in massive concrete elements is being developed, taking account of state-dependent properties of the hardening concrete. This contribution focuses on the evolution of the thermal characteristics of early-age concrete, as a function of the state of the hardening process. A literature review is presented, together with new test results obtained by means of improved test methods.

Specific heat of hardening concrete

Literature review

According to a 1982 state-of-the-art report,⁴ published data on the thermal characteristics of hardening concrete appear to be scarce. The available results dealing with the evolution of specific heat during hardening are now summarized chronologically.

In 1946, Löfqvist⁴ found a minor decrease of the specific heat of cement mortar during hardening — from

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1.18 kJ/kg per K at an age of 3 days to 1.16 kJ/kg per K at 10 days. In 1970, Brown and Javard⁵ determined a more fundamental decrease. They stated that the specific heat of hardening concrete decreases linearly with time by about 20%. Maréchal,^{6,7} in 1972 and 1973, found only minor fluctuations of the specific heat during hardening. The linear decline with time found by Brown and Javard was refuted in 1982 by Reinhardt *et al.*⁸ who, for hardening concrete, proposed a linear function of the degree of hydration also with a reduction of 20%

$$c = c_0(1.25 - 0.25r) \quad (1)$$

where c is the specific heat of the hardening concrete, c_0 is the specific heat of the hardened concrete and r is the degree of hydration.

However, Reinhardt *et al.* do not give the origin of this relation. In 1982 a test method for the specific heat during hardening, taking account of the hydration heat production, was described by Hansen *et al.*⁹ For cement paste they found a decline from 1.92 kJ/kg per K at 1.5 to 1.65 kJ/kg per K at 120 h. The decline was found to be linear with the logarithm of time, which for common cement types is very similar to a linear decline with degree of hydration. In his 1988 research on thermal stresses in hardening concrete, Hamfler,¹⁰ like many others, neglected any fluctuations of the specific heat during hardening. In 1991, Van Breugel¹¹ postulated a linear decrease with degree of hydration, based on the fictitious specific heat of the hydrated cement

$$c = \frac{1}{\rho} [G_c c_{cf} r + G_c(1 - r) + G_a c_a + G_w c_w] \quad (2)$$

where G_c , G_a , G_w are the amounts by weight of cement, aggregate and water, c_c , c_a , c_w are the specific heats of cement, aggregate and water, and c_{cf} is the fictitious specific heat of the hydrated part of cement. This implies that

$$c_{cf} = 0.0084\theta + 0.339 \text{ (kJ/kg per K)} \quad (3)$$

where θ is the actual temperature of the sample.

Almost every author in the literature review reported a decrease of the specific heat during hardening, while nobody found an increase. However, there seems to be no agreement concerning the magnitude of the decrease. It is significant that some researchers published data obtained on concrete specimens, while others tested cement paste or mortar. When all the results are

converted, in an indicative way, towards values for a conventional gravel concrete consisting of 300 kg cement, 150 kg water, 670 kg sand and 1280 kg gravel, the decreases summarized in Table 1 are found.

Determination of the specific heat of hardening cement paste

Test method As mentioned above, Hansen *et al.*⁹ proposed a test method for the determination of the specific heat of hardening cement paste which is very similar to the traditional methods. The specific heat is determined by measuring the temperature rise in a calorimeter caused by a certain energy supply. The first measurement is taken in the absence of the cement paste and serves as a calibration of the calorimeter. A known energy supply E_1 causes a temperature increase $\Delta\theta_1$. A second measurement takes place in the presence of the cement paste, leading to a temperature increase $\Delta\theta_2$ caused by an energy supply E_2 . It can be proved that the specific heat of the cement paste c_p is given by

$$c_p = \frac{1}{m_p} \left(\frac{E_2}{\Delta\theta_2} - \frac{E_1}{\Delta\theta_1} \right) \quad (4)$$

where c_p is the specific heat of the cement paste, m_p is the mass of the cement paste sample, E_2 and E_1 are the energy supplies with and without the paste sample respectively, and $\Delta\theta_2$ and $\Delta\theta_1$ are the temperature rises with and without the paste sample respectively.

At the Magel Laboratory this principle was used in order to study the evolution of specific heat as a function of the degree of hydration. Fig. 1 shows the test equipment. The calorimeter is placed in a thermostatic waterbath and is partly filled with heptane. Heptane is used instead of water because of its lower specific heat (± 2100 J/kg per K). This enables a more accurate determination of the specific heat of the paste sample. The energy E_1 or E_2 is supplied by means of a known electrical resistance. The effect of the heat of hydration is taken into account implicitly by determining the temperature rise $\Delta\theta_2$ as shown in Fig. 2. The temperature history in the calorimeter is recorded during a sufficiently long period before and after the supply of energy. The slope of the linear parts of the curves before and after the supply of energy is determined by two effects, namely the heat exchange with the surrounding waterbath and the heat production due to hydration. The temperature rise $\Delta\theta_2$ is determined by the intersection points A and B of the

Table 1. Evolution of thermal characteristics of concrete during hardening

	Ref. 4	Ref. 5	Refs 6, 7	Ref. 8	Ref. 9	Ref. 11	The present study
Specific heat c	-1%	-20%	Constant	-20%	-5%	-4%	-13%
Thermal diffusivity a	Constant	-15%	Maximum at 50 h		+12% (paste)		-9%
Thermal conductivity λ		-30%	Maximum at 50 h	-50%			-21%

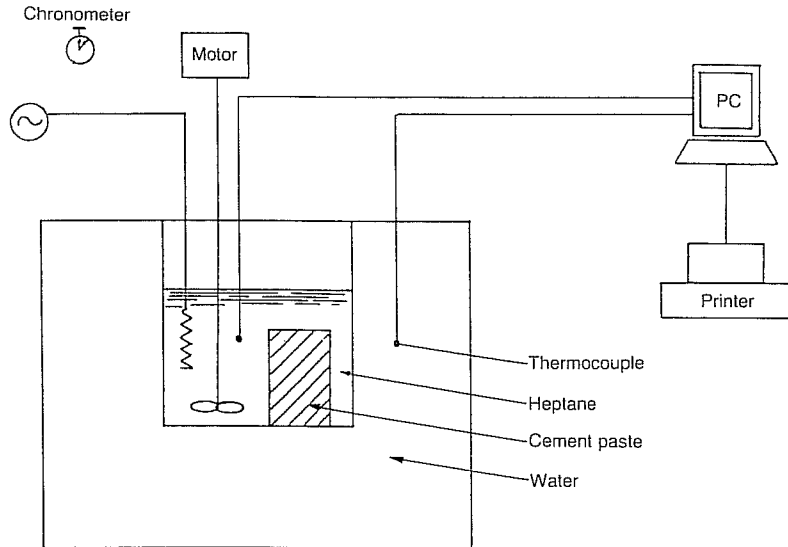
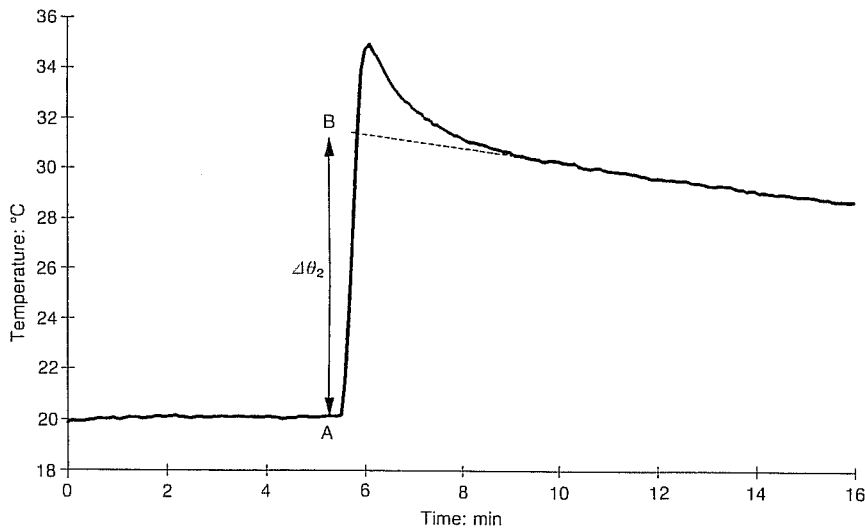


Fig. 1 Calorimeter

Fig. 2. Determination of the temperature rise $\Delta\theta_2$

extrapolation of the linear parts with the almost vertical part of the recorded temperature history. The temperature peak immediately after the supply of energy is due to the fact that a certain time is needed to reach thermal equilibrium between the cement paste and the heptane.

Test results The tests were carried out on a cement paste with water–cement ratio (w/c) of 0.5, using blastfurnace slag cement CEM III/C 32.5. In order to obtain the evolution of the specific heat during hardening, cement paste samples were tested at ages ranging from 1 day to 7 days. The corresponding degree of hydration was calculated as the fraction of heat released, using a newly developed hydration model.² Until testing, the cement paste samples were stored at 20°C and 95% relative humidity.

The test results are shown in Fig. 3. A linear decrease

with degree of hydration holds quite well. Linear regression yields

$$c_p(r) = 1300(1.5 - 0.5r) \text{ (J/kg per K)} \quad (5)$$

For the fresh cement paste ($r = 0$) this yields $c_p = 1950$ J/kg per K, which coincides very well with the value $c_p = 1954$ J/kg per K obtained by applying a superposition rule on a cement paste with $w/c = 0.5$ and specific heats of the cement and water of 857 and 4187 J/kg per K.

Equation (5), valid for cement paste, can be converted into an expression for the conventional gravel concrete mentioned above:

$$c(r) = 1000 (1.5 - 0.15r) \text{ (J/kg per K)}$$

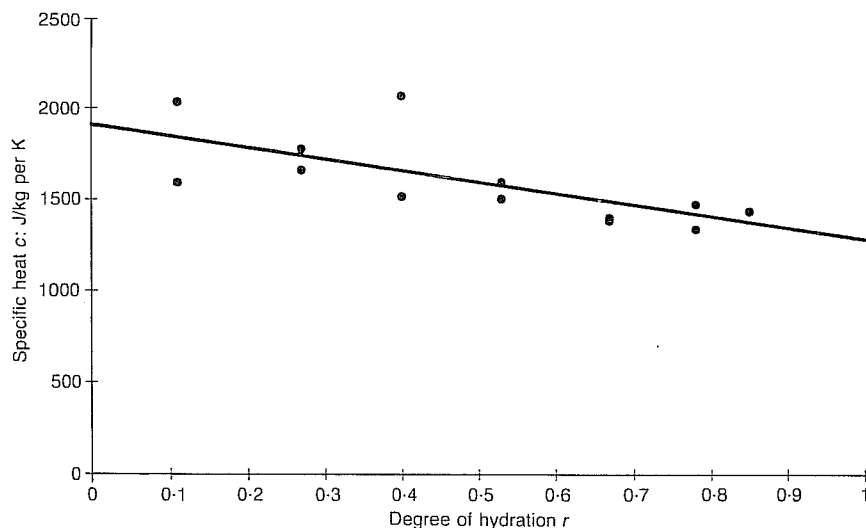


Fig. 3. Specific heat of hardening cement paste CEM III/C 32.5: test results

This result shows a decrease of the specific heat of concrete during hardening of about 13%.

Discussion

Considering the conversion of free water into chemically bound water, a decrease of the specific heat during hardening of concrete was to be expected. The decrease obtained of about 13% is more or less intermediate in terms of the literature results (Table 1). The differences in Table 1 might be explained by different test circumstances, e.g. paste, cement or concrete; w/c; cement type; aggregate type; humidity.

Thermal diffusivity of hardening concrete

Literature review

While for specific heat there is some fundamental agreement between the various researchers, literature data on the thermal diffusivity of hardening concrete do not seem to be coherent. Constant values during hardening are reported as well as increasing and decreasing tendencies. In 1946, Löfquist⁴ reported a constant diffusivity for concretes with different compositions. For example, a concrete with 350 kg cement and w/c = 0.5 had a constant thermal diffusivity $a = 0.0037 \text{ m}^2/\text{h}$ for ages ranging from 1 to 7 days.

Brown and Javaid⁵ also published test results for the period ranging from 1 to 7 days. For hardening concrete, they found a decrease from $0.0034 \text{ m}^2/\text{h}$ at 1 day to $0.0029 \text{ m}^2/\text{h}$ at 7 days. Maréchal^{6,7} postulated that the thermal diffusivity reaches a maximum at an age of about 50 h. His test results are based on concrete with rapid hardening Portland cement, using a very simple test procedure (see below). By means of the same test method, Hansen *et al.*⁹ found an increase of the thermal diffusivity of hardening cement paste from about 0.85 to about $1.05 \text{ m}^2/1000 \text{ h}$. The evolution of the thermal

diffusivity during hardening according to the various researchers is summarized in Table 1. The values are for tests on concrete, unless otherwise stated.

Determination of the thermal diffusivity of hardening concrete

Test method For thermal diffusivity measurements, an experimental non-stationary method was devised by Maréchal^{6,7} and by Hansen *et al.*⁹ In this method a cylindrical specimen at temperature θ_0 is suddenly immersed in a waterbath at temperature $\theta_0 + \Delta\theta$. The temperature $\theta(t)$ at the cylinder axis is measured as a function of time. The thermal diffusivity a can then be calculated by fitting the experimental results to a theoretical expression based on a Bessel function of the first kind of order zero.⁹ This fitting is done by computer calculations. To avoid these difficult calculations another fitting approach can be applied. When $\log[(\theta_0 + \Delta\theta - \theta(t))/\Delta\theta]$ is plotted as a function of time, the curve becomes linear after some time, as shown in Fig. 4. The slope of the experimental line thus obtained is directly related to the thermal diffusivity.

Hansen proposed a $\Delta\theta$ of 2 K. It can be shown² by numerical simulation that for concrete specimens the heat of hydration then influences the results adversely. In the case of cement paste, as for example in Hansen's test, the influence of the heat of hydration is expected to be even worse. Less influence is noticed when $\Delta\theta = 5 \text{ K}$ or even 10 K.

Test results The thermal diffusivity of hardening concrete was determined on sealed cylindrical concrete specimens of diameter 100 mm, made with concrete having the following composition per m^3 : 300 kg blastfurnace slag cement CEM III/C 32.5, 150 kg water, 670 kg sand, 1280 kg gravel. The cylinders are permanently stored at 20°C . During the test $\Delta\theta = 10 \text{ K}$ is applied. Tests were carried out at ages ranging from

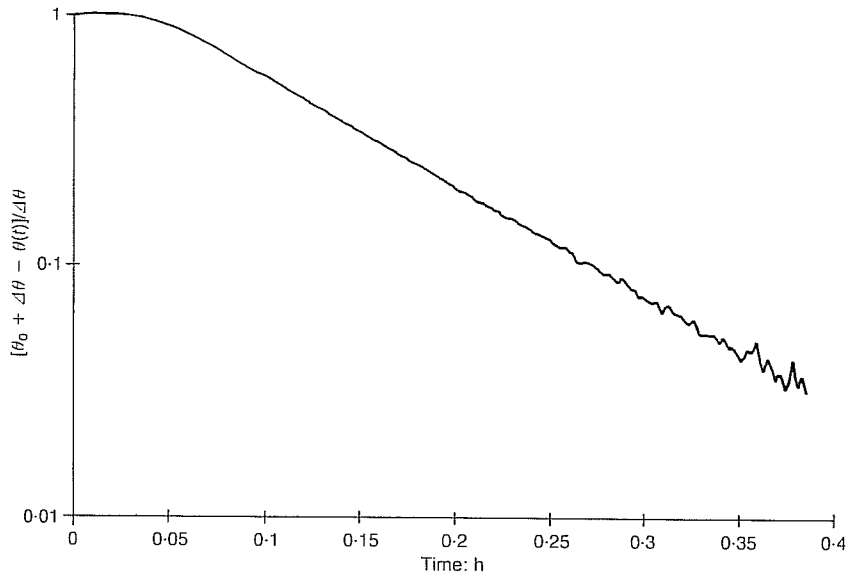


Fig. 4. Temperature history at cylinder axis during a diffusivity test

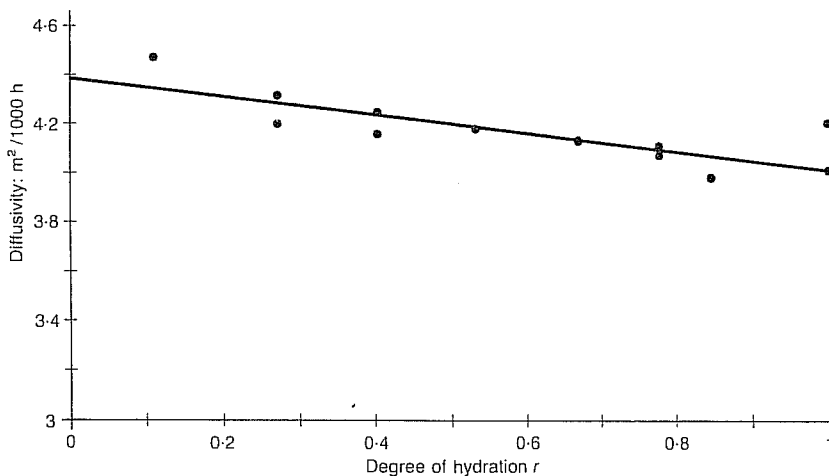


Fig. 5. Thermal diffusivity of hardening concrete CEM III/C 32.5: test results

day to 14 days. Fig. 4 shows the recorded temperature history at the cylinder axis during a diffusivity test at an age of 3 days.

The values obtained for the thermal diffusivity of hardening concrete are shown in Fig. 5 as a function of the degree of hydration r . It can be concluded that a linear relation agrees very well with the test results. Linear regression leads to

$$a(r) = 4 \times 10^{-3}(1.10 - 0.10r) \text{ (m}^2\text{/h)} \quad (6)$$

For the thermal diffusivity, there is no mixing rule for the conversion of concrete data into paste data or vice versa.

Discussion

Based on the diffusivity tests on hardening concrete, a decrease during hardening of about 9% is found. In Table 1 this value is compared with literature results. Why Hansen reports an increase (on cement paste) is not clear.

Physically a decrease is expected because of a decrease in free water and an increase in porosity due to hydration. As mentioned above, the heat production of the cement paste might have influenced Hansen's test results. The initial increase found by Maréchal, however, is not likely to have been caused by the heat of hydration effect. Maréchal used very small cylinders (diameter 72 mm) and low test temperatures (0°C) combined with a $\Delta\theta$ of 10 K. However, he tested concrete with 350 or 500 kg rapid hardening Portland cement.

Indications for the evolution of the thermal diffusivity during hardening can also be obtained via a combination of specific heat and thermal conductivity:

$$a = \frac{\lambda}{\rho c} \quad (7)$$

where λ is thermal conductivity and ρ is the density of the concrete (= constant).

Table 1 gives values for the evolution of the thermal

conductivity of concrete according to several researchers. Brown and Javai⁵ determined the conductivity experimentally, and found a decrease of about 30%. Reinhardt *et al.*⁸ postulated a decrease of 50%, but without showing how this figure is derived. As the conductivity seems to decrease more than the specific heat, a decrease in the diffusivity is expected.

Conclusions

1. For the determination of the specific heat of cement paste during hardening, a test method was developed based on the method of Hansen *et al.*⁹
2. The specific heat was determined on hardening cement paste samples made with blastfurnace slag cement CEM III/C 32.5. It was concluded that the specific heat decreases linearly with the degree of hydration.
3. For the determination of the thermal diffusivity of concrete during hardening, a test method was developed based on the method of Maréchal^{6,7} and Hansen *et al.*⁹
4. By means of tests on hardening concrete cylinders it was concluded that the thermal diffusivity decreases linearly with increasing degree of hydration.

Appendix. Calculation of the degree of hydration

The degree of hydration $r(t)$ of a hardening cementitious material is defined as the fraction of cement that has already hydrated:

$$r(t) = \frac{\text{amount of cement that has reacted at time } t}{\text{total amount of cement at time } t = 0} \quad (8)$$

A practical method to estimate the degree of hydration is based on the exothermic character of the hydration process:

$$r(t) = \frac{Q(t)}{Q_{\max}} \quad (9)$$

where $Q(t)$ is the total heat developed at time t (in J/g) and $Q_{\max}$ is the total heat development corresponding to complete hydration (in J/g).

The total heat $Q(t)$ can be calculated as

$$Q(t) = \int_0^t q(t) dt \quad (10)$$

where $q(t)$ is the heat production (in J/g per h).

In Ref. 2 a new hydration model is developed, yielding a heat production q depending on the actual degree of hydration $r(t)$ and the actual temperature $\theta(t)$

$$q(t) = q[r(t), \theta(t)] \quad (11)$$

From equations (10) and (11) it follows that at time t

$$\frac{dQ(t)}{dt} = q(t) = q[r(t), \theta(t)] \quad (12)$$

Differentiation of equation (9) and the introduction of equation (12) yield

$$dr(t) = \frac{dQ(t)}{Q_{\max}} = \frac{q[r(t), \theta(t)]}{Q_{\max}} dt \quad (13)$$

Hence, the value of r can be updated by substituting equation (13) in

$$r(t + dt) = r(t) + dr(t) \quad (14)$$

In this way the age of a specimen continuously stored at 20°C can be transformed into the corresponding degree of hydration. For more details on the hydration model $q[r(t), \theta(t)]$, which is to be determined experimentally, see Ref. 2.

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